Lecture 6: Likelihood Ratio Test

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Suggested Reading: Hogg et al. 2018, Chapter 8.8.

1 Testing the goodness of fit, Likelihood ratio test, Score test and Wald test

Suppose data $X_1, X_2, \cdots, X_n$ are independent and generated from pmf/density $f(x;\theta)$, i.e., the density and pmf is controlled by the parameter θ. This section aims at significant testing, i.e., testing

$$H_0: \ \theta = \theta_0, \text{ where } \theta_0 \text{ is a known number}$$

Consider the log-likelihood function

$$\ell(\theta) = \sum_{i=1}^{n} \log(f(X_i, \theta)), \ \widehat{\theta} = \arg\max_{\theta \in \mathbf{R}} \ell(\theta)$$

Under the null, $\sqrt{I(\theta_0)}(\widehat{\theta} - \theta_0) \to_d N(0,1)$. Based on this asymptotic property, we are able to derive a good hypothesis testing procedure.

Theorem 1.1. *Suppose data $X_1, X_2, \cdots, X_n$ are independent and generated from pmf/density $f(x;\theta)$, i.e., the density and pmf is controlled by the parameter $\theta \in \mathbb{R}$ (scalar parameter). For testing*

$$H_0: \ \theta = \theta_0, \ \textit{here } \theta_0 \textit{ is a known number}$$

Define the test statistics

$$\widehat{T} = 2(\ell(\widehat{\theta}) - \ell(\theta_0)) \tag{1}$$

then under H_0,

$$\widehat{T} \to_d \chi_1^2$$

Lemma 1.2. *Suppose data $X_1, \cdots, X_n$ with joint density $f(x_1, \cdots, x_n, \theta)$, then the Fisher information is given by*

$$I(\theta) = -\mathbf{E}\left[\frac{\partial^2}{\partial \theta^2} \log(f(X_1, \ldots, X_n, \theta))\right]$$

Proof [An informal proof to Theorem 1.1] From Taylor expansion,

$$\ell(\theta_0) \approx \ell(\widehat{\theta}) + \frac{\mathrm{d}\ell(\widehat{\theta})}{\mathrm{d}\theta}(\theta_0 - \widehat{\theta}) + \frac{1}{2}\frac{\mathrm{d}^2\ell(\widehat{\theta})}{\mathrm{d}\theta^2}(\theta_0 - \widehat{\theta})^2$$
$$\Rightarrow 2(\ell(\widehat{\theta}) - \ell(\theta_0)) \approx -\frac{\mathrm{d}^2\ell(\widehat{\theta})}{\mathrm{d}\theta^2}(\theta_0 - \widehat{\theta})^2 \approx (\sqrt{I(\theta_0)}(\widehat{\theta} - \theta_0))^2$$

Since under the null $\sqrt{I(\theta_0)}(\widehat{\theta} - \theta_0) \to_d N(0,1)$, thus $2(\ell(\widehat{\theta}) - \ell(\theta_0)) \approx Z^2$ with $Z \sim N(0,1)$, which is χ_1^2. □

Remark If $\theta \neq \theta_0$, then $\ell(\theta) \approx \ell(\widehat{\theta}) > \ell(\theta_0)$, so the test statistics should be positively large. Based on this observation, we choose rejection region $R_\alpha = [\chi_{1,\alpha}, \infty)$ with $\chi_{1,\alpha}$ the α upper quantile of χ_1^2 distribution, and the P-value is given by

$$\widehat{p} = 1 - \Psi(\widehat{T}), \text{where } \Psi \text{ is cdf of } \chi_1^2 \text{ distribution.}$$

Similar to the likelihood ratio test, we can establish another hypothesis testing procedure, called the **Wald test**, based on the asymptotic property mentioned above.

$$I(\widehat{\theta}) := -\frac{\mathrm{d}^2\ell(\widehat{\theta})}{\mathrm{d}\theta^2}$$

and thus we also have

$$\widehat{W} = I(\widehat{\theta})(\widehat{\theta} - \theta_0)^2 \to_d \chi_1^2$$

Example 1 Suppose data $X_1, \cdots, X_n$ satisfy normal distribution $N(\mu, \sigma^2)$ with unknown μ and known σ^2. We want to test the hypothesis

$$H_0 : \mu = \mu_0 \text{ versus } \ H_1 : \mu \neq \mu_0$$

The log-likelihood function is given by

$$\ell(\mu) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 - \frac{n}{2}\log(2\pi\sigma^2)$$

and the MLE is $\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$. So the likelihood ratio test is

$$\widehat{T} = 2(\ell(\widehat{\mu}) - \ell(\mu_0)) = \frac{n(\widehat{\mu} - \mu_0)^2}{\sigma^2} \to_d \chi_1^2$$

The $P - value$ is

$$\widehat{p} = 1 - \Psi(\widehat{T}),$$

here $\Psi(\cdot)$ is the cdf of χ^2_1 distribution.

Furthermore, note that

$$I(\widehat{\mu}) = -\frac{\mathrm{d}^2\ell(\widehat{\mu})}{\mathrm{d}\mu^2} = \frac{n}{\sigma^2},$$

then the Wald test is based on:

$$\frac{n(\bar{X}-\mu_0)^2}{\sigma^2} \to_d \chi^2_1.$$

The rejection region is still $R_\alpha = [\chi^2_{\alpha,1}, \infty)$.

Example 2 Suppose data $X_1, \cdots, X_n$ are i.i.d. with exponential distribution, i.e., with density

$$f(x, \lambda) = \lambda \exp(-\lambda x),$$

and we want to test the null hypothesis

$$H_0 : \lambda = \lambda_0 \text{ versus } H_1 : \lambda \neq \lambda_0.$$

The log–likelihood function is $\ell(\lambda) = n\log(\lambda) - \lambda\sum_{i=1}^n X_i$, whose derivative is $\frac{n}{\lambda} - \sum_{i=1}^n X_i$, and MLE is $\widehat{\lambda} = \frac{n}{\sum_{i=1}^n X_i}$. From this, the likelihood ratio test statistics is

$$\widehat{T} = 2(\ell(\widehat{\lambda}) - \ell(\lambda_0)) = 2n\log(\frac{\widehat{\lambda}}{\lambda_0}) - 2\sum_{i=1}^n X_i(\widehat{\lambda} - \lambda_0).$$

Since $\ell''(\lambda) = -\frac{n}{\lambda^2}$, the Wald test statistics is

$$\frac{n}{\widehat{\lambda}^2}(\widehat{\lambda} - \lambda_0)^2.$$

Finally, we introduce another test based on the score function, called the **score test**. Recall the definition of a score function

$$U(\theta) = \frac{\partial}{\partial\theta}\log(f(X_1, \cdots, X_n, \theta)).$$

Theorem 1.3. *Suppose data $X_1, X_2, \cdots, X_n$ are independent and generated from pmf/density $f(x;\theta)$, i.e., the density and pmf is controlled by the parameter $\theta \in \mathbb{R}$. For testing*

$$H_0 : \ \theta = \theta_0, \ \textit{here } \theta_0 \textit{ is a known number}$$

Define the test statistics

$$\widehat{T} = \frac{U^2(\theta_0)}{I(\theta_0)} \tag{2}$$

then under H_0,

$$\widehat{T} \to_d \chi_1^2$$

Proof [An informal proof to Theorem 1.3] Still, from Taylor's theorem

$$U(\theta_0) \approx U(\widehat{\theta}) + \frac{\partial U(\widehat{\theta})}{\partial \theta}(\theta_0 - \widehat{\theta}) \approx I(\theta_0)(\widehat{\theta} - \theta_0) \tag{3}$$

so

$$\frac{U^2(\theta_0)}{I(\theta_0)} \approx [\sqrt{I(\theta_0)}(\widehat{\theta} - \theta_0)]^2 \to_d \chi_1^2$$

□

Example 3 Similarly, suppose $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$, then the score function is

$$U(\mu) = \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu),\ I(\mu) = \frac{n}{\sigma^2}$$

so the test statistics is

$$\widehat{T} = \frac{n(\widehat{\mu} - \mu_0)^2}{\sigma^2} \to_d \chi_1^2$$

2 Likelihood Ratio Test for composite null

We generalize the above likelihood ratio type tests to general settings, where we may have **composite null hypotheses** and composite alternative hypotheses.

A likelihood ratio test (LRT) statistic for testing $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ is

$$\lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta|x)}{\sup_{\theta\in\Theta} L(\theta|x)} = \frac{\sup_{\theta\in\Theta_0} L(\theta|x)}{\max\{\sup_{\theta\in\Theta_0} L(\theta|x), \sup_{\theta\in\Theta_1} L(\theta|x)\}}$$

where $\Theta = \Theta_0 \cup \Theta_1$. A LRT accepts H_0 if $\lambda(x)$ is large enough, and rejects H_0 is $\lambda(x)$ is too small, i.e., the rejection region of LRT is $\{x : \lambda(x) \le c\}$ for some $c \in [0, 1]$.

Note that the LRT can be written as a function of MLEs: denote $\widehat{\theta}_0 = \arg\max_{\theta\in\Theta_0} L(\theta|x)$ as the (restricted) MLE within the set Θ_0, and denote $\widehat{\theta} = \arg\max_{\theta\in\Theta} L(\theta|x)$ as the (unrestricted) MLE within the whole set Θ, thus we have the LRT statistics is

$$\lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta|x)}{\sup_{\theta\in\Theta} L(\theta|x)} = \frac{L(\widehat{\theta}_0|x)}{L(\widehat{\theta}|x)}.$$

Note that $-\log\lambda(x) = \ell(\hat{\theta}) - \ell(\hat{\theta}_0)$, which equals to the test statistics in Theorem 1.1 when H_0 is $\theta = \theta_0$. Thus, the above LRT statistics can be viewed as a generalization of the test in equation (1) to the case of composite null hypothesis (However, the limiting distribution is not necessarily χ^2_1, e.g., see example 4 below).

Example 3: testing normal mean Consider $X_1, \ldots, X_n$ a random sample from $N(\mu, \sigma^2)$, where σ^2 is known.

$$H_0 : \mu = \mu_0, \quad vs. \quad H_1 : \mu \neq \mu_0.$$

In this case, we have $\Theta_0 = \{\mu_0\}$ and $\Theta_1 = (-\infty, \mu_0) \cup (\mu_0, +\infty)$, thus $\widehat{\theta}_0 = \mu_0$ and $\widehat{\theta} = \bar{X}$. Thus

$$\begin{aligned}\frac{L(\widehat{\theta}_0)}{L(\widehat{\theta})} &= \exp\left(-\frac{1}{2\sigma^2}[\sum(X_i - \mu_0)^2 - \sum(X_i - \bar{X})^2]\right) = \exp\left(-\frac{1}{2\sigma^2}[2n\bar{X}(\bar{X} - \mu_0) + n\mu_0^2 - n\bar{X}^2]\right) \\ &= \exp\left(-n\frac{1}{2\sigma^2}(\bar{X} - \mu_0)^2\right)\end{aligned}$$

Thus $\lambda(x) \leq c \iff |\bar{X} - \mu_0| \geq k$ for some value k. Thus the LRT for this problem has the rejection region

$$R(\alpha) = \{\frac{|\bar{X} - \mu_0|}{\sigma/\sqrt{n}} \geq z_{\alpha/2}\}.$$

with significance level α.

If the variance σ^2 is unknown, then we have $\Theta_0 = \{\mu_0\} \times (0, \infty)$ and $\Theta_1 = ((-\infty, \mu_0) \cup (\mu_0, +\infty)) \times (0, \infty)$, The MLEs are $\widehat{\theta}_0 = (\mu_0, \frac{1}{n}\sum(X_i - \mu_0)^2)$ and $\widehat{\theta} = (\bar{X}, \frac{1}{n}\sum(X_i - \bar{X})^2)$. Thus

$$\lambda(x) = \frac{L(\widehat{\theta}_0)}{L(\widehat{\theta})} = \left(\frac{\sum(X_i - \bar{X})^2}{\sum(X_i - \mu_0)^2}\right)^{n/2} = \left(\frac{1}{1 + \frac{n(\bar{X} - \mu_0)^2}{\sum(X_i - \bar{X})^2}}\right)^{n/2}$$

thus $\lambda(x) \leq c \iff \frac{n(\bar{X} - \mu_0)^2}{\sum(X_i - \bar{X})^2} \geq k \iff \frac{\sqrt{n}|\bar{X} - \mu_0|}{S} \geq k'$, where $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is the sample variance. Thus the rejection region with significance level α is

$$R(\alpha) = \{\frac{\sqrt{n}|\bar{X} - \mu_0|}{S} \geq t_{\alpha/2, n-1}\}.$$

Remark (One-side alternative situation) Now we suppose $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$. By definition the likelihood ratio test statistics is given by

$$\lambda(x) = \left(\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu_0)^2\right)\right) / \left(\exp\left(\sup_{\mu \in [\mu_0, \infty)} -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2\right)\right)$$

For $H(\mu) = \frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \mu)^2$ is decreasing on $(-\infty, \widehat{\mu}]$ and increasing on $[\widehat{\mu}, \infty)$, we have

$$\widehat{T} = \begin{cases} 1 \text{ if } \widehat{\mu} < \mu_0 \\ \left(\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \mu_0)^2\right)\right) / \left(\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \widehat{\mu})^2\right)\right) \text{ otherwise} \end{cases}$$

Example 4: composite null Again testing normal mean, $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with σ^2 known. Consider testing

$$H_0 : \mu \leq \mu_0 \text{ versus } H_1 : \mu > \mu_0$$

In such cases, we have $\Theta_0 = (-\infty, \mu_0]$ and $\Theta_1 = [\mu_0, +\infty)$, $\Theta = \mathbb{R}$. Therefore we have $\widehat{\theta}_0 = \min\{\bar{X}, \mu_0\}$, $\widehat{\theta} = \bar{X}$, and thus the likelihood ratio statistics is

$$\lambda(x) = \frac{L(\widehat{\theta}_0)}{L(\widehat{\theta})} = \begin{cases} 1 & \text{if } \bar{X} \leq \mu_0 \\ \exp\left(-n\frac{1}{2\sigma^2}(\bar{X} - \mu_0)^2\right) & \text{if } \bar{X} > \mu_0 \end{cases}$$

Therefore, the LRT will reject H_0 when $\lambda(x) \leq c$, which happens iff $\bar{X} > \mu_0$ and $\bar{X} - \mu_0 \geq k$, thus the rejection region is

$$R(\alpha) = \{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\}.$$

Furthermore, we note that under the null hypothesis, if the true mean value is $\mu < \mu_0$, the corresponding type-I error

$$\beta(\mu) = \mathbf{P}\{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\} = \mathbf{P}\{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} + \frac{\mu - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha\} \leq \alpha$$

Which means that for composite null hypothesis, the type-I error is always controlled by α.

Example 5: testing normal variance Consider $X_1, \ldots, X_n$ a random sample from $N(\mu, \sigma^2)$. Assume both mean and variance are unknown. Test:

$$H_0 : \sigma = \sigma_0, \quad vs. \quad H_1 : \sigma \neq \sigma_0.$$

In this case, we have $\Theta_0 = \mathbb{R} \times \{\sigma_0^2\}$ and $\Theta_1 = \mathbb{R} \times ((0, \sigma_0^2) \cup (\sigma_0^2, +\infty))$, and MLEs are $\widehat{\theta}_0 = (\bar{X}, \sigma_0^2)$ and $\widehat{\theta} = (\bar{X}, \frac{1}{n}\sum(X_i - \bar{X})^2)$. Thus

$$\frac{L(\widehat{\theta}_0)}{L(\widehat{\theta})} = \left(\frac{\frac{1}{n}\sum(X_i - \bar{X})^2}{\sigma_0^2}\right)^{n/2} \exp\left\{-\frac{1}{2\sigma_0^2}\sum(X_i - \bar{X})^2 + \frac{n}{2}\right\}$$

Thus $\lambda(x) \leq c$ happens when $S = \frac{\frac{1}{n}\sum(X_i - \bar{X})^2}{\sigma_0^2}$ is $S \leq c_1$ or $S \geq c_2$, where c_1, c_2 are chosen by controlling the significance level α.

Theorem 2.1 (Wilks)**.** *Suppose the dimension of Θ_0 is q and the dimension of Θ is r, define*

$$\lambda_n = 2\log\left(\frac{\sup_{\theta\in\Theta}\mathcal{L}_n(\theta)}{\sup_{\theta\in\Theta_0}\mathcal{L}_n(\theta)}\right) = 2\log\left(\frac{\mathcal{L}_n(\hat{\theta})}{\mathcal{L}_n(\hat{\theta}_0)}\right),$$

where $\hat{\theta}$ is the MLE and $\hat{\theta}_0$ is the MLE when the parameter space is constrained to Θ_0.

Then under the null hypothesis,

$$\lambda_n \xrightarrow{d} \chi^2_{r-q},$$

where χ^2_{r-q} denotes the χ^2 distribution with the degree of freedom being $r-q$.

Moreover, the p-value for the test is $\mathbf{P}(\chi^2_{r-q} \geq \lambda_n)$.

Example 6: Mendel's peas Gregor Mendel bred peas of four types of progeny: $\{green, yellow\} \times \{round, wrinkled\}$. The number of peas in each type follows a multinomial distribution with the probability vector $p \equiv (p_1, p_2, p_3, p_4)$ denoting the probabilities for round yellow, wrinkled yellow, round green, and wrinkled green, respectively. Denote the numbers of the peas in each progeny as $X \equiv (X_1, X_2, X_3, X_4)$, then

$$X \sim MN(p).$$

His theory of inheritance predicts that the probability vector is $p_0 \equiv (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$. The same information is presented in the table below.

		color	
		yellow	green
shape	round	9/16	3/16
	wrinkled	3/16	1/16

Table 1: H_0 for the four types of progeny

The hypothesis testing problem is formulated to test

$$H_0 : p = p_0 \text{ v.s. } H_1 : p \neq p_0.$$

To perform the likelihood ratio test, we need to pin down the dimension difference between the unconstrained parameter space and the null hypothesis parameter space, and compute the likelihood ratio test statistic λ_n with MLEs.

The unconstrained parameter space is $\Theta = \{p : \sum_{i=1}^4 p_i = 1, p_j \in [0,1] \text{ for } j = 1, ..., 4\}$ and $dim(\Theta) = 3$ due to the equality constraint; $\Theta_0 = \{p = p_0\}$ and $dim(\Theta_0) = 0$ as Θ_0 only contains one point. Therefore, by Wilks' theorem, under the null hypothesis,

$$\lambda_n \xrightarrow{d} \chi^2_3.$$

Therefore, we can perform the asymptotically size α test:

$$\text{reject } H_0 \text{ if } \lambda_n \geq \chi^2_{3,\alpha}$$

where $\chi^2_{3,\alpha}$ denotes the upper α quantile of χ^2_3. Additionally, the p value of the likelihood ratio test is approximately $\mathbf{P}(\chi^2_3 \geq \lambda_n)$.

To compute the test statistic, we need to find the MLE of p. The MLE of p is the empirical proportion:

$$\hat{p} = X/n.$$

With data $X = (315, 101, 108, 32)$ and $n = 556$, we can compute the test statistic:

$$\lambda_n = 2\log\left(\frac{\prod_{i=1}^4 \hat{p}_i^{X_i}}{\prod_{i=1}^4 p_{0,i}^{X_i}}\right) = 2\sum_{i=1}^4 X_i \log\left(\frac{X_i/n}{p_{0,i}}\right) \approx 0.48.$$

With the $\alpha = 5\%$ chosen before the experiment, we retain the null hypothesis. Additionally, the p value is approximately 0.92.

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*